

nOdes

OBSTRUCTION MANUAL

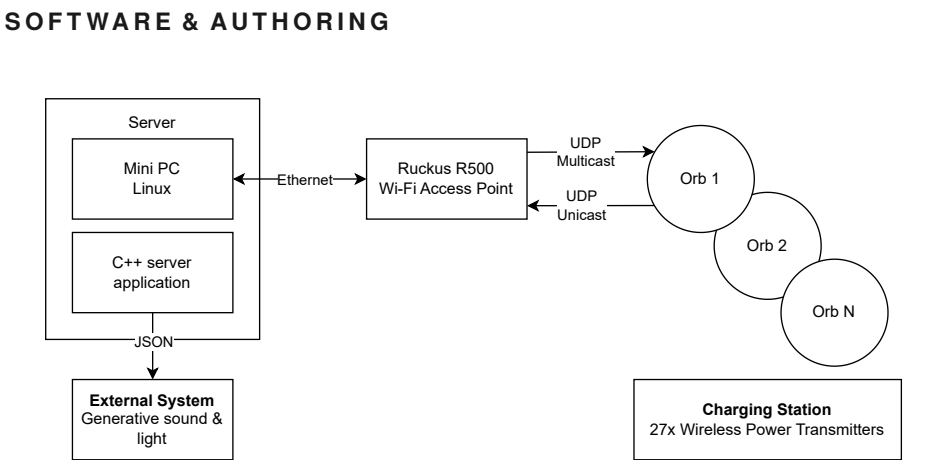
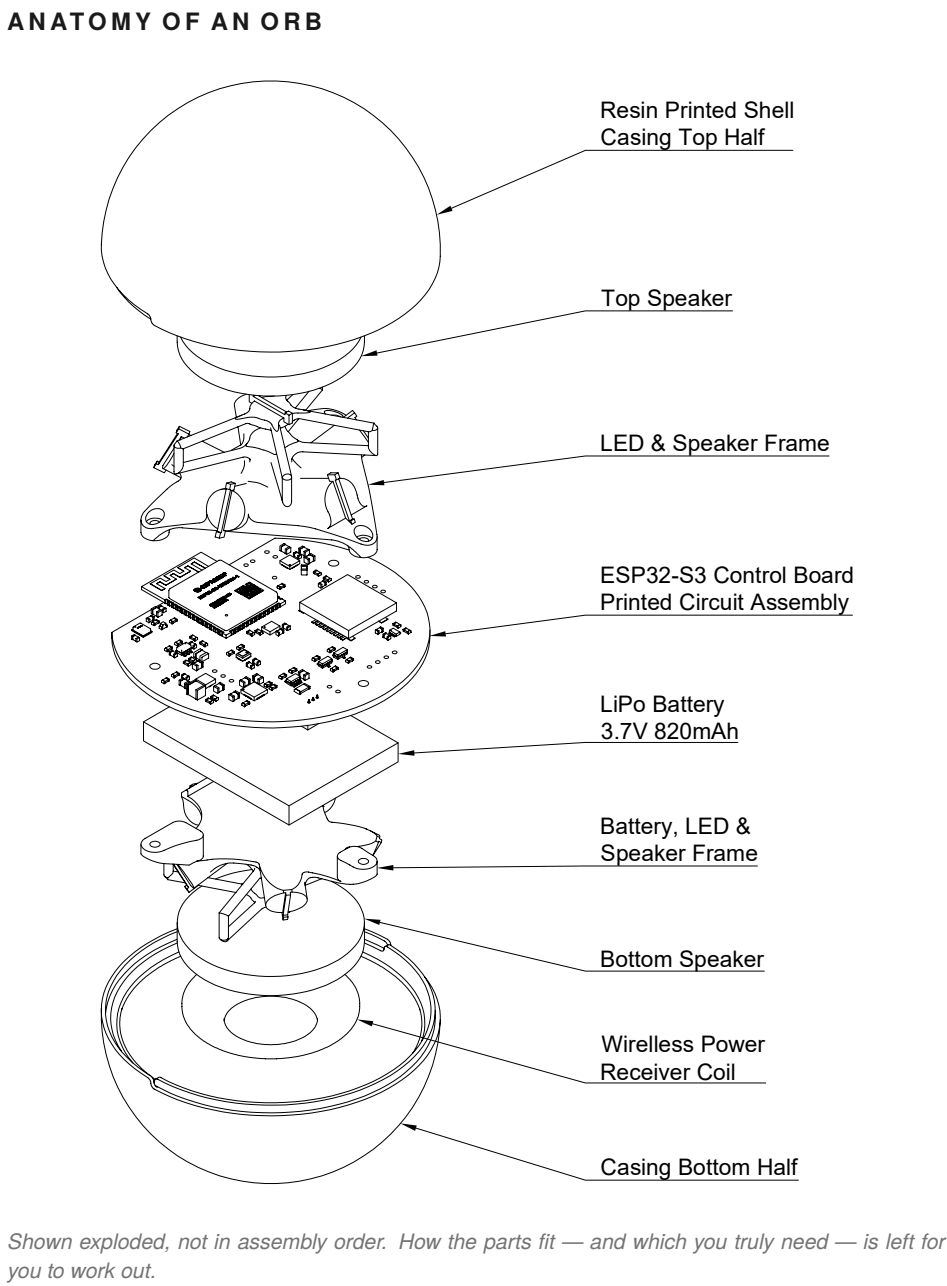
*an open intuitive score
for a swarm of light*

This is not a step-by-step guide for replication, but a prompt for reinterpretation. Open it fully as a poster, or fold it to A5. The reverse is left blank — for your own notes.

THE INSTRUMENT

A constellation you play together

nOdes is a swarm of spherical, handheld *orbs* — each a self-contained wireless device with motion sensing, twelve icosahedrally-arranged LEDs, audio and battery power. Ten to twenty-four people play it at once, shaping a shared harmonic environment by rotating, shaking, passing and clustering their orbs. The swarm responds to the crowd — not the other way around.



Orbs stream motion to a server over Wi-Fi at 50 Hz. A C++ server reduces the whole swarm to a few collective measures and hands them on as light and sound.

Authoring an experience

The orbs stay sealed — their firmware travels wirelessly, *over the air*, so you rarely open one. But the art lives on the *server*. A new piece is a short **Python** script over the live state of the whole swarm: read motion and clustering, *prompt* the crowd (“find your colour”, “make one group”), measure how the collective responds, and answer in light and sound on the orbs.

Omitted by design: the exact packet layout, slot allocation and firmware. Rediscover them, or ask the repository.

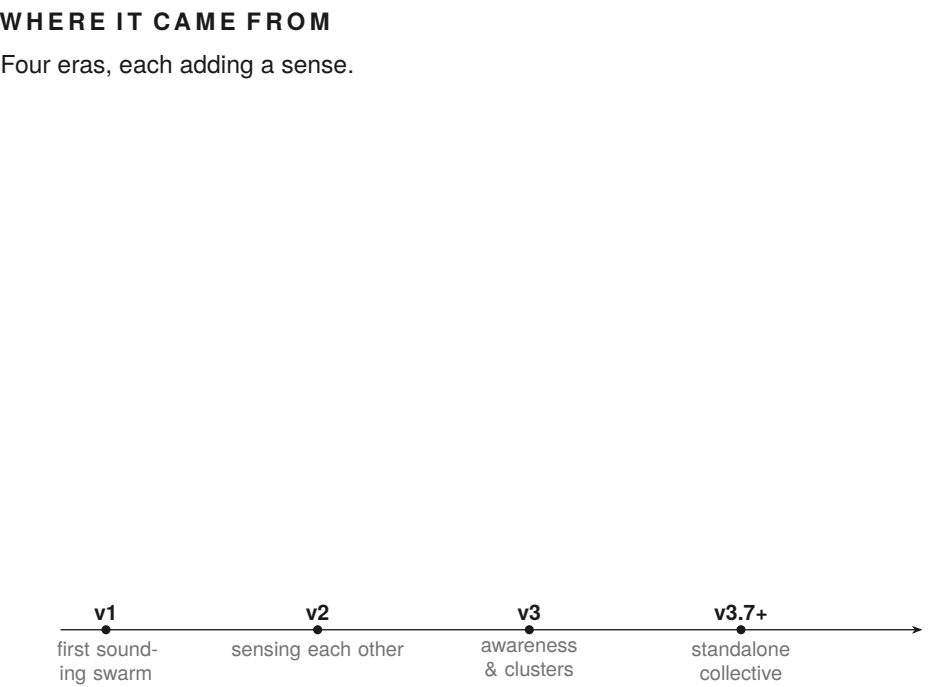
PHENOMENOLOGY OF THE SWARM

The swarm is a single shifting body made of many hands.

No one plays a melody; everyone tends a weather. Structure and chaos take turns. You feel part of a collective before you understand how.

The feedback loop runs from sensor to collective metric to light and sound on the orbs themselves — not through loudspeakers in the room. It grows out of swarm robotics and community music-making: an instrument that needs other people.

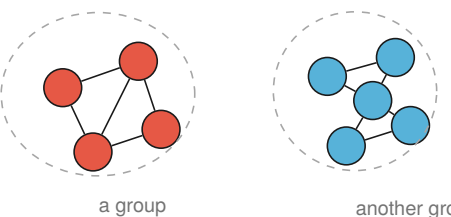
This manual is intentionally incomplete. Following the Obstruction Manual method, it withholds steps to leave room for your own making. The gaps are the invitation.



First sound; then mutual awareness; then collective behaviour; then the resilience to keep playing when the network falls quiet.

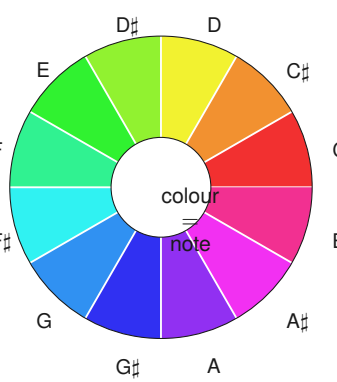
1 · CLUSTERING

The swarm finds its groups



Bring orbs close and they form a group. The swarm works out its own clusters from who is near whom — and *each group takes on a colour*. This is where the music begins, so start here.

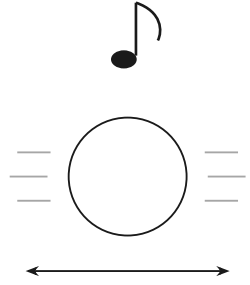
2 · COLOUR = NOTE



Each colour is a note. A group glows its colour and sounds that note — so a *different cluster sings a different note*. Twelve colours, twelve notes, around the wheel.

3 · SHAKE = CHIME

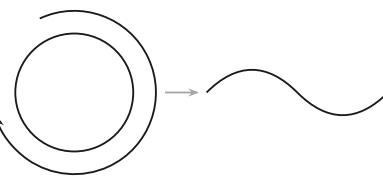
Shake an orb



A sharp shake rings a *chime* — a bright, percussive transient that sounds over the cluster’s note. The harder the swarm moves, the more it sparkles.

4 · SPIN = PAD

Spin an orb



Keep an orb turning and its note swells into a *sustained pad* — a long, held voice. Hold the motion, hold the sound; let it fall still and the pad fades.

WHEN IT RESISTS

An orb won’t join? It is waiting for the access point. Power-cycle it close to the AP.

One orb glows the wrong colour in a big swarm? Past about twenty-five orbs the marginal orb starves of downlink. It is coverage, not a fault — move the access point, or thin the crowd.

Silence at rest? That is correct. The swarm sings on movement and change, not on stillness.

CREDITS

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